

IDEA League

Master of Science in Applied Geophysics
Research Thesis

From Particles to Waves

**Towards coupling Dynamic Snow Modelling and Spectral Elements to
detect Avalanche Breakoff in DAS Data.**

Salome Anna Bachmann

Thursday July 2, 2026

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Master of Science Thesis

for the degree of Master of Science in Applied Geophysics
by

Salome Anna Bachmann

Thursday July 2, 2026

IDEA LEAGUE
JOINT MASTER'S IN APPLIED GEOPHYSICS

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Abstract

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I would like to thank all the supervisors involved in this thesis.

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Introduction

Snow avalanches are a major natural hazard in mountainous regions; they endanger human life as well as infrastructure (Schweizer et al., 2003). The Swiss Institute for Snow and Avalanche Research has reported an average of 22 fatalities caused by avalanches every year, with the number of non-fatal accidents being tenfold higher (SLF, 2025). These data show that avalanches are a significant threat, which is why it is crucial to understand how avalanches form and propagate in order to protect people and infrastructure better.

Avalanches are rapid masses of snow that move down mountain slopes due to gravity (Ancey, 2001). Snow avalanches can either be loose snow, which starts from a point in a cohesionless surface, wet snow, or snow slab avalanches (Schweizer et al., 2003). In this Master Thesis, the focus will be on dry snow slab avalanches, so mostly coherent masses of snow which contain no liquid water. Snow slab avalanches consist of a cohesive slab of snow (Schweizer et al., 2003) that is released because of loading at the surface, which triggers crack propagation in a weak layer (Gaume et al., 2017). Weak layers are characterized by the grain shape of snow, which leads to lower cohesion of those layers (Föhn et al., 1998).

As mentioned, avalanches are a major threat to human life in mountainous regions, which is why detecting the breakoff of the snow slab is important for hazard mitigation. Detecting the breakoff of the slab means that warnings can be issued in lower parts of the mountain that an avalanche is possible. Furthermore, detecting slab breakoff with DAS is also important to further understand crack dynamics in snow as well as the factors influencing crack propagation.

In this master's thesis, the aim is to detect avalanche slab breakoff and crack propagation using modeled data from distributed acoustic sensing (DAS) cables.

In this thesis, a fibre-optic cable installed on the slope is treated as a continuous seismic sensor, so that deformation can be observed along its full length rather than at only a few isolated stations (Parker et al., 2014). This is possible because DAS sends optical pulses through the fibre and records the backscattered light from every position along the cable (Shatalin et al., 1998). Ground motion near the cable, for example weak-layer failure or slab breakoff, changes the fibre strain, which appears as phase variations in the backscattered signal (Hartog, 2017). The phase variations are then converted into dynamic strain and from strain, into deformation along the cable (Trabattoni et al., 2023). mention also the cable location here, length etc

A pilot early-warning system using distributed acoustic sensing has been implemented in Norway, where avalanches are detected by their seismic signature (Turquet et al., 2024), meaning when the avalanche is already moving downhill. Such sensors are also located in Vallée de la Sionne in the

Swiss Alps (Paitz et al., 2023). The detectability of avalanches by DAS has also been studied in the Swiss Alps by Paitz et al., 2023 and Edme et al., 2023, but the detectability of the slab breakoff itself has not been studied.

A multitude of models are used to study different processes in snow. The SNOWPACK and CROCUS models used in the Swiss and French alps model the structure and evolution of snow (Brun et al., 1992; Bartelt and Lehning, 2002). These models were developed to understand how snow evolves under different conditions, for example a changing temperature gradient and water content and what consequences this has for the formation of weak layers where failure likely occurs and avalanches are triggered (Brun et al., 1992; Bartelt and Lehning, 2002). SNOWPACK and CROCUS are comprehensive models that take into account multiple governing equations and processes and what consequences they have for the structure of snow. While it is important to understand the influences on for example temperature gradient on the snow structure and weak layer formation and stability as a consequence, this is not the focus of the thesis. This is because these models are too complex and comprehensive for the purpose of trying to detect slab breakoff in DAS data as they model the entire snow structure. Using SNOWPACK or CROCUS to investigate how different snowpack compositions and factors influence the DAS signal is beyond the scope of this thesis.

To model crack propagation in snow, multiple numerical modeling methods have been used. Cundall and Strack, 1979 developed the distinct element method (DEM) for granular media. This model discretizes the domain as spheres and disks, which interact with each other based on Newton's second law (Cundall and Strack, 1979). The interactions of particles are modeled by the governing equations and the resulting forces and displacements (Cundall and Strack, 1979). This method has been used by Bobillier et al., 2020; Bobillier et al., 2024; Mulak and Gaume, 2019 and many others to model crack propagation and its influencing factors. The main limitation of DEM is that the particle contacts and the corresponding equations need to be recalculated every time a grain contact is made or broken (Cundall and Strack, 1979). This means that for large, slope-scale simulations, this method becomes increasingly computationally expensive. In addition to the high computation cost, DEM also cannot account for anticrack propagation, the closure of cracks in this propagation mode is accounted for by manually removing elements from the mesh, which is not very efficient (Gaume et al., 2019).

The same is true for finite element modelling (FE or FEM), which is done using SALVUS in this Thesis. FEM has been used to model snow mechanics since the 1970s, a review of the different approaches has been published by Podolskiy et al., 2013. The finite element modelling approach used in this thesis is described more in detail in ?? . FE can also not account for anticrack propagation, similarly to DEM, mesh elements would need to be removed to account for anticrack propagation.

This problem is solved by using the material point method (MPM). In this method, the material is discretized in a set of Lagrangian points, which are used to track mass, deformation and momentum (Bardenhagen et al., 2000). Because the points are not fixed, an Eulerian background mesh is needed to calculate spatial derivatives (Bardenhagen et al., 2000). This method is not only more computationally efficient than DEM on the slope-scale, it can also account for anticrack behavior and strain softening of snow (Gaume et al., 2019). This is why the material point method will also be used in the thesis to investigate slab breakoff in DAS data on a large scale. The material point method has been used by Bergfeld et al., 2025, who modeled slope-scale experiments to understand the transition of crack propagation modes, Trottet et al., 2022 who modeled propagation saw tests to observe transitions in crack propagation speeds, as well as Gaume et al., 2019 and Gaume et al., 2018.

While MPM is very good at accounting for anticrack propagation, FE is more computationally efficient and accurate Lian et al., 2012. This is relevant especially for large-scale simulations, such as the ones done in this Thesis. This is because in MPM, the discretization through particles causes the displacement of each particle to be tracked and the contact at each particle to be calculated (Zhang

et al., 2023). Furthermore, MPM does not include boundary conditions in the way FE does (Coombs et al., 2025). Finite element is more efficient on the slope scale because the mesh is fixed, as described by (Zienkiewicz et al., 2025). This means that the degrees of freedom and thus the number of calculations which need to be done are much smaller than in the MPM case. This means that MPM is more accurate at representing snow as a medium with its porous character and anticrack behavior, but FE is more efficient at performing large-scale simulations.

While it is important to understand the different modeling methods and their limitations, it is also important to understand the different types of crack propagation and what factors influence them. Understanding how fast a crack can propagate and which factors influence this is important for warning, as it relates the time of the initial crack to the time where the slab breaks off depending on snow and terrain parameters.

Two main types of crack propagation are believed to be important for weak layer failure: anticrack propagation and shear crack propagation (Bergfeld et al., 2025). There are three ways in which cracks can open up: mode I, where space is opened between the crack faces under tensile loading, and modes II and III, which represent parallel shearing and diagonal shearing (Heierli et al., 2003). In anticrack propagation, the weak layer consolidates under loading (C. A. Knight and N. C. Knight, 1972), while cracks propagate in a combination of modes I-III (Heierli et al., 2003). Gaume et al., 2018 modeled anticrack propagation induced by a propagation saw test (PST) using MPM, which showed that the scale of the simulations influences the propagation direction of the cracks. For smaller scale simulations, cracks propagate from top to bottom of a slope, whereas for slope-scale simulations, cracks propagate bottom to top, which means that remote triggering of avalanches is possible (Gaume et al., 2018). This is an important consideration for the thesis because it influences where the slab breakoff would be observed in terms of DAS cable location. Other models also show that the crack propagation speed of anticracks is influenced by slope angle and curvature: concave slope areas have higher propagation speeds than convex areas of the slope (Gaume et al., 2019). Furthermore, there is a directionality in propagation speed, simulations by Gaume et al., 2019 show that cracks propagate twice as fast upslope than in the downslope direction, which is especially important when thinking about remote triggering. The slab breaks off when the slope angle exceeds the friction angle of the material (Gaume et al., 2019). Slope angle also influences which mode of crack propagation is more dominant (Rosendahl and WeiSSgraeber, 2020). In steeper terrains, shear modes (II and III) dominate, whereas in flatter terrain, mode I dominates the crack propagation (Rosendahl and WeiSSgraeber, 2020). The critical force, which is for example exerted by a skier, needed for weak layer failure is also dependent on slope angle; it decreases with increasing slope angle (Rosendahl and WeiSSgraeber, 2020). Snow parameters also have an influence on this critical force, as shown by simulations by Rosendahl and WeiSSgraeber, 2020: the thicker the slab or the higher the slab tensile strength, the better the force by the skier can be distributed on the weak layer and the more force is needed for failure.

Multiple experiments and numerical models have shown that cracks can propagate faster than the elastic wave speed in snow, so-called supershear crack propagation (Trottet et al., 2022; Bobillier et al., 2024; Bergfeld et al., 2025). Understanding what causes the transition from sub-Rayleigh to supershear crack propagation is of high importance for the thesis, because it can give an estimate of whether supershear crack propagation will occur under certain conditions, which leads to faster propagation of cracks and less time between loading and slab breakoff. In MPM models of PST, Trottet et al., 2022 observe that the occurrence of supershear cracks is dependent on slope angle: in steeper terrains, supershear propagation was observed. This can be explained by slab bending, because in steep terrains, bending of the slab is not limited to the collapse depth of the weak layer (Trottet et al., 2022). In in steeper terrain, the slab can bend more because of the terrain angle, the additional bending increases tensile stress in the slab, which leads to faster crack propagation (Trottet

et al., 2022). These observations were also made by Bobillier et al., 2024 as well as Bergfeld et al., 2025. The release area of an avalanche is also tied to propagation speed: at supershear propagation speed, there is less crack arrest and therefore larger release areas (Trottet et al., 2022; Bergfeld et al., 2025). This is crucial because larger release areas also mean a larger snow volume, which has more potential for damage.

Independent of crack propagation mode, different factors influence the speed and distance at which cracks can propagate in the weak layer before failure occurs (Gaume et al., 2018; Gaume et al., 2017; Gaume et al., 2015). As discussed above, slope angle and stiffness are important factors. Other factors include weak layer parameters: the weaker the weak layer, the slower crack propagation; the thicker the weak layer, the lower the crack propagation distance before failure (Gaume et al., 2015). The depth of the slab also has an influence on propagation distance and speed: the thicker the slab, the larger the propagation distance and velocity (Gaume et al., 2015). How fast a slab will break off is also dependent on the critical crack length needed for failure (Gaume et al., 2017). If the snowpack is more stable, which is the case when the slab has a high elastic modulus or the weak layer is thicker, the critical crack length is higher (Gaume et al., 2017) because the snow can sustain longer cracks before it fails. A very important factor in material strength is loading rate: if loading is slow, sintering can occur (Mulak and Gaume, 2019). This means that new snow contacts can form, which can strengthen the material (Mulak and Gaume, 2019). This is in contrast to rapid loading, which happens for example due to a skier (Mulak and Gaume, 2019).

In order to understand how crack propagation and slab breakoff would look in DAS data, numerical modeling of the wave propagation is done. This is done in SALVUS, which is a package that uses spectral elements to model seismic wavefields (Afanasyev et al., 2017).

The different snow processes described have different implications for the seismic wavefield, which will be modeled. Firstly, the loading which occurs before crack nucleation changes the stress state of the snowpack. The weak layer collapse and anticrack propagation produce volumetric compaction in the snowpack, this produces a seismic signal. The crack propagation, whether sub-Rayleigh or supershear, creates a moving rupture front, which can be likened to a moving source of seismic waves, the cracks itself also create a seismic signal. Lastly, the slab breakoff and sliding creates a seismic signal because a large mass of snow moves downhill, which likely creates the largest seismic signal of all of the processes mentioned. The signals of the different processes described will be modeled in the thesis, which will help to understand how the different processes look in DAS data and whether they can be distinguished from each other.

Based on the above information, there are multiple considerations necessary in order to model DAS data of crack propagation accurately. The scale of the models is an important factor, because as mentioned, it influences the direction of crack propagation. This influences at which location along or relative to the DAS cable slab breakoff will occur, which changes the measured data. Furthermore, multiple parameters influence the time passed between initial cracking and slab breakoff, these are important to consider when interpreting data. The different modes of crack propagation are also important to consider, it is possible that a transition between sub-Rayleigh and supershear crack propagation can be observed in the data. The loading rate can also be considered, because slower loading leads to stabilizing processes such as sintering.

This thesis will focus on the modelling of crack propagation, as recorded by DAS. Sub-Rayleigh as well as supershear crack propagation will be modelled using different methods and models. The aim is to determine what the different cases of crack propagation will look like in DAS data. This is then relevant for detecting actual avalanches in real-world data. The main questions are what crack propagation in different regimes looks like, how it differs when different modeling methods are used, and how the results can be explained in terms of wave physics.

Theory

2.1 Spectral ElementWave Propagation Modeling

The model described in ?? is implemented using the finite-element (FE) method in SALVUS. The following section describes the physics used by SALVUS. Understanding the physics behind the modeling has relevance for understanding and interpreting the results, but also what can be modeled. To discretize a domain, the domain is divided into a finite number of elements (Zienkiewicz et al., 2025). The number of elements in this case is defined by the maximum wave frequency that needs to be resolved accurately, the number of mesh elements per wavelength is user-input. The boundaries of these elements describe the mesh, the mesh has a finite number of grid-points (edge points of the elements) (Zienkiewicz et al., 2025). The governing equations are determined at every element node,

so for rectangular elements with nodes 1,2,3 and 4, there is the system $r^1 = \begin{pmatrix} r_1^1 \\ r_2^1 \\ r_3^1 \\ r_4^1 \end{pmatrix}$ for force and

$u^1 = \begin{pmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \\ u_4^1 \end{pmatrix}$ for displacement (Zienkiewicz et al., 2025). The residual or characteristic relationship of

linear elastic elements is of the form $r^1 = f^1 - K^1 u^1$, where f are the nodal forces distributing load on the nodes and K is the stiffness matrix of a specific element (Zienkiewicz et al., 2025). Because in FE, the approximation of the wavefield by basis functions leads to a representation over the whole domain and not just at a set of points (Afanasiev et al., 2019). This means integrals can be calculated over surfaces or even volumes (Afanasiev et al., 2019).

Snow is a porous medium (Mellor, 1977), which is approximated as a solid in this Thesis. This is because SALVUS meshes need an isotropic material and cannot account for pore spaces. Higher frequencies in the resulting data would be attenuated much quicker in an actual porous medium (Koushik et al., 2024). In addition, the presence of pore spaces in the medium would lead to complex scattering of the fields in the results, their absence is an important factor to consider when comparing simulated data to real-world data. The presence of air or water in snow is not neglected completely in the SALVUS simulations implemented however, because the material properties taken from Guillemot et al., 2021 are based on field measurements of snow. This means that the velocities measured by Guillemot et al., 2021 are not of pure snow, but snow that contains air or water.

The governing equation considered in this case is the elastic wave equation, because real-life crack

propagation has P- and S-waves. The wave equation for a heterogenous elastic solid in the n -dimensional domain G is

$$\rho \partial_t^2 u = \nabla \cdot \sigma + f \quad (2.1)$$

(Afanasiev et al., 2019; Aki and Richards, 1981). Stress can be related to displacement using Hoek's law: $\sigma = \frac{C}{\nabla u}$, where C is a 4th order tensor which characterizes the stiffness of the medium (Afanasiev et al., 2019). Combining ?? with Hoek's law, this leads to

$$\rho \partial_t^2 u = \nabla \cdot \frac{C}{\nabla u} + f, \quad (2.2)$$

which is the strong form of the linear elastic wave equation.

Boundary conditions, such as free surfaces, as well as initial conditions must be stated explicitly, they have the form $\sigma \cdot \hat{n}|_{\partial G} = \bar{\tau}(x), u|_{t=0} = \bar{u}(x), v|_{t=0} = \bar{v}(x)$ (Afanasiev et al., 2019). In this case, $\bar{\tau}(x), \bar{u}(x), \bar{v}(x)$ are the spatially variable distributions of traction, displacement and velocity (Afanasiev et al., 2019). The outward pointing normal vector on the boundary of domain G is described by $\hat{n}|_{\partial G}$.

FE methods use the weak form of the governing equations (Zienkiewicz et al., 2025). To obtain the weak form of the linear elastic wave equation in ??, the whole equation needs to be multiplied with a test function w , which is not time-dependent (Afanasiev et al., 2019). The weak form enforces the solution of the governing equation across the domain, instead of solving the equation everywhere, an integration over time leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_G w \cdot (\nabla \cdot \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (2.3)$$

(Afanasiev et al., 2019). The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in an enclosed volume: $\iiint_V (\nabla \cdot F) dV = \oint_S (F \cdot \hat{n}) dS$, where F is the vector field. When this is applied to the first term on the right in ??, this leads to

$$\int_G \rho w \cdot \partial_t^2 u \, d^n x = \int_{\partial G} w \cdot (\sigma \cdot \hat{n}) \, d^{n-1} x - \int_G (\nabla w : \sigma) \, d^n x + \int_G w \cdot f \, d^n x \quad (2.4)$$

(Afanasiev et al., 2019). The first integral on the left side of ?? is the boundary integral, which changes depending on the boundary conditions implemented. The integral on the right hand side is the mass term, the second on the left hand side is the stiffness term, and the last on the left hand side is the external forcing term, all in ???. Equation ?? can be compacted to

$$(\rho \partial_t^2 u, w)_G - \langle \sigma \cdot \hat{n}, w \rangle_{\partial G} + (\sigma, \nabla w)_G = (f, w)_G, \quad (2.5)$$

using the notations $(a, b)_G = \int_G a \cdot b \, d^n x$, $\langle a, b \rangle_{\partial G} = \int_{\partial G} a \cdot b \, d^{n-1} x$ as well as Hoek's law (Afanasiev et al., 2019). This can be even more compacted by describing a mathematical model I with M for mass term, B for boundary term, S for stiffness and F for forcing term:

$$I_I(M) - I_I(B) + I_I(S) = I_I(F) = \sum_{P \in \{M, B, S, F\}} I_I(P) = 0 \quad (2.6)$$

(Afanasiev et al., 2019).

To solve ?? the Galerkin approach is used, the test function w and solution u are approximated by ϕ , which are a set of functions from the same space (Afanasiev et al., 2019). Any discrete function, for example displacement can be approximated this way:

$$g(x) \approx \sum_{j=1}^N F_j \phi_j(x), \quad (2.7)$$

where F_j is the expansion of the basis and g is the continuous function which is the approximation (Afanasiev et al., 2019). The derivatives of a function can then be approximated by the derivatives of ϕ .

To evaluate the integrals in ??, numerical quadrature is used Afanasiev et al., 2019. A certain number of integration weights W are combined, $g(x)$ and the derivative of $g(x)$ are evaluated at a set of integration points x_i (Afanasiev et al., 2019). This results in

$$\int_G g(x) d^n x \approx \sum_{i=1}^M W_i^G \sum_{j=1}^N F_j \phi_j(x_i) \quad (2.8)$$

(Afanasiev et al., 2019). In FE, an element-specific reference coordinate system θ is used. In this thesis, the domain consists of a snow layer and an air layer, which means there is a sharp discontinuity in model parameters. As Afanasiev et al., 2019 state, the model parameters and strain field are not differentiable. This is why a smooth polynomial basis of the governing equations over the whole domain is not appropriate, the basis B needs to consist of piecewise functions G defined on the individual elements (Afanasiev et al., 2019). The integration weights and basis function derivatives used, according to Afanasiev et al., 2019, are pre-tabulated in the element-specific coordinate system θ . To transform the integrals and derivatives to physical coordinates x , so the coordinates of the snow extent in this case, the element-specific Jacobian $\frac{\partial x}{\partial \theta}$ is used (Afanasiev et al., 2019). Equations ?? and ?? and their derivatives become:

$$g(x) \approx \sum_{j=1}^N F_j \phi_j(\theta(x)) \quad (2.9)$$

and

$$\nabla g(x) \approx \sum_{j=1}^N F_j \frac{\partial}{\partial \theta} \phi_j(\theta(x)) \frac{\partial \theta}{\partial x} \quad (2.10)$$

(Afanasiev et al., 2019). In equations ?? and ??, the function space V is approximated as $V_h \approx V$ in a previous step, this means the accuracy of the approximation is dependent on how well the original function space is approximated (Afanasiev et al., 2019). To make calculations over the whole domain G , the contributions from each element need to be summed:

$$\int_{G_e} g(x) d^n x \approx \sum_{i=1}^M W_i^{G_e} \sum_{j=1}^N F_j \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^n \quad (2.11)$$

for the volume and

$$\int_{\partial G_e} g(x) d^{n-1} x \approx \sum_{i=1}^M W_i^{\partial G_e} \sum_{j=1}^N F_j \phi_j(\theta(x_i)) \left| \frac{\partial x}{\partial \theta} \right|_{x=x_i}^{n-1} \quad (2.12)$$

for the surface of the domain (Afanasiev et al., 2019).

The dependence on the reference coordinate system is described by E , which is a model containing the physical coordinate to reference coordinate transformation, in other words, the determinant of the Jacobian matrix and its inverse (Afanasiev et al., 2019). Using basis B of the polynomial functions of the approximation V_h , which are piecewise per-element because of possible discontinuities, the abstract form in ?? becomes

$$\sum_{P \in \{M, B, S, F\}} I_I(P, B(E)) = 0 \quad (2.13)$$

(Afanasiev et al., 2019). This means that fields such as displacement and velocity are calculated in this way. From these fields, the data from the model in section ?? is obtained and plotted.

2.2 From Particles to Waves: Translating MPM to SE

The material point simulations of the PST performed by [NEED TO CITE WHO MADE THE MPM HERE](#) result in the recorded stresses in the model according to Gaume et al., 2019. This results in the stretch factor ϵ comprised of the material properties C_{ij} such as Young's modulus, Lamé parameters and Poisson ratio as well as the stress matrix σ :

$$\epsilon = C_{ij}\sigma \quad (2.14)$$

(holzapfelNonlinearSolidMechanics2010). The equation can be rewritten to obtain the stresses in different directions, in the xx direction for example:

$$\sigma_{xx} = (+2\mu)\epsilon_{xx} + \lambda\epsilon_{yy} \quad (2.15)$$

(Aki and Richards, 1981). When scaling this with the particle volume, the individual moment tensor components can be obtained using

$$M_{ij} = \sigma_{ij}V_p. \quad (2.16)$$

These moment tensor components can then be input into SALVUS as custom source time functions. How are the waveforms of the stfs obtained

Methodology

3.1 Model Setups

3.1.1 Simple Model

The same two-layer medium as in section ?? was set up. In the simple case, a series of SALVUS moment tensor sources are set up with a certain distance. The time delay with which the sources fire is dependent on the sub-rayleigh or supershear case which is implemented. The bedrock or soil is not considered in both models. This is because preliminary tests of a 3-layered medium have shown that it acts as a transmitter of waves. For computational efficiency, the bedrock layer was thus neglected and the bottom of the snow was made a regular boundary, in other words, reflecting, to mimick the presence of a layer below the snow.

3.1.2 Cohesive Zone Model

A graphical representation of the model setup and the equations used can be found in appendix ??.

The rupture model used in the simulations is implemented as a mixed-mode Cohesive Zone Model (CZM) following Mahajan and Joshi, 2008. This model was chosen because there is a narrow band ahead of the fracture zone, the cohesive zone, where the material is already weakened. This means that before a point in the material cracks, there is already weakening of the material before. This seems to be reasonable for representing crack propagation in snow, since failure occurs as a consequence of loading and is not spontaneous. The computational domain extends 400 m in the horizontal (x) direction and 3 m in the vertical (y) direction and contains a 1.5 m snow layer over a 1.5 m air layer. Material parameters for the snow layer are taken from Guillemot et al., 2021. The mixed-mode behavior is implemented because in reality, crack propagation can involve both normal and shear displacement.

The mesh is generated to resolve frequencies up to the 95th percentile of the source wavelet's frequency content. Cohesive tractions are prescribed on zero-thickness interface elements and depend on the normalized opening $\xi = \frac{\delta}{\delta_c}$. The zero-thickness interface elements are implemented by not solving for volumetric strain but displacement between interfaces instead. For the linear softening law used here the traction in a given cohesive element follows

$$T(\xi) = T_{\text{peak}} (1 - \xi) \quad \text{for } 0 \leq \xi \leq 1, \quad T(\xi) = 0 \text{ for } \xi > 1,$$

where T_{peak} is the peak cohesive traction and δ_c the critical opening. The normalized opening can be related to the local crack coordinate as $\xi = \frac{(x-x_0)}{l_{\text{crack}}}$ for a crack front with initial position x_0 and local

process length l_{crack} .

The mixed-mode potential used in the code follows the form from Mahajan and Joshi, 2008. With $\Delta = (\Delta_n, \Delta_t)$ denoting normal and tangential separations, the potential is written as

$$\phi(\Delta) = \phi_n \exp\left(-\frac{\Delta_n}{\delta_n}\right) \left\{ \left[1 + \frac{\Delta_n}{\delta_n}\right] + q \left[1 - \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right)\right] \right\},$$

from which the normal and tangential tractions are obtained, the implementation uses the same closed-form expressions. This is so that peak tractions occur at $\Delta_n = \delta_n$ and $|\Delta_t| = \delta_t/\sqrt{2}$. The normal and shear works of separation relate to peak strengths via

$$\phi_n = e \sigma_{\max} \delta_n, \quad \phi_t = \sqrt{\frac{e}{2}} \tau_{\max} \delta_t,$$

and the critical openings for pure modes are computed as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \quad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

with the mixed-mode critical opening

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2}.$$

Lamé parameters are derived from input seismic speeds and density: $v_s = \sqrt{\frac{\mu}{\rho}}$ so that $\mu = \rho v_s^2$, and $v_p = \sqrt{\frac{(\lambda+2\mu)}{\rho}}$ so that $\lambda = \rho(v_p^2 - 2v_s^2)$. Young's modulus is then computed from the Lamé parameters: $E = \frac{\mu(3\lambda + 2\mu)}{\lambda + \mu}$ (Aki and Richards, 1981). These conversions are applied for every material in the model.

The critical opening distances for pure normal (Mode I) and shear (Mode II) separations are computed from the corresponding fracture energies, strenghts, and peak tractions as

$$\delta_{n,c} = \frac{2G_I}{\sigma_n}, \quad \delta_{s,c} = \frac{2G_{II}}{\tau_s},$$

and combined for mixed-mode behaviour using

$$\delta_c = \sqrt{(1 - r_{\text{mode}}) \delta_{n,c}^2 + r_{\text{mode}} \delta_{s,c}^2},$$

where $r_{\text{mode}} \in [0, 1]$ controls the mode I/II mix, where 0 is pure mode I and 1 is pure mode II. In the current implementation the slip on each subfault follows the prescribed linear slip law up to the critical slip distance. The mode mix is an user-input parameter, while all the other parameters are derived from laboratory experiments from Mahajan and Joshi, 2008.

The characteristic Irwin length for shear is calculated as $l_{ch} = \frac{E_{\text{eff}} G_{II}}{\tau_s^2}$, and the cohesive zone length is taken as $l_{cz} = \frac{\pi}{8} l_{ch}$, which sets the spatial extent over which cohesive weakening is active ahead of the crack tip.

Rupture propagation speed is chosen with respect to the local shear-wave speed and the bounds discussed in Trottet et al., 2022. Where appropriate, the rupture speed is corrected for finite slab thickness and bending stiffness to account for slab flexure effects. The rupture front is discretized into a sequence of subfaults: each subfault is activated at an onset time

$$t_o = \frac{x - x_{\text{start}}}{v_{\text{rupture}}},$$

using the subfault horizontal position x , the rupture starting position x_{start} , and the prescribed rupture speed v_{rupture} .

Each activated subfault is represented as a point/source in the SALVUS source array. The scalar seismic moment for a subfault is computed from the shear modulus, the prescribed slip, and the subfault width Δx :

$$M_0 = \mu x_{\text{slip}} \Delta x.$$

Moment-tensor components are set according to the user-specified mode-mix ratio, e.g. here the implementation uses $m_{xx} = 0$, $m_{xy} = M_0 r_{\text{mode}}$, $m_{yy} = -(1 - r_{\text{mode}})M_0$, which yields pure mode behaviour at the extremes of $r_{\text{mode}} = 0$ or 1 , which means that an additional component of the moment tensor source becomes 0 . All subfault sources use the same central frequency, wavelet shape, and amplitude scaling so the rupture is represented as a coherent moving source in the wave propagation simulations.

Key parameters and their provenance used in the simulations are summarised below:

Parameter	Typical value / source
σ_n	0.00184 MPa from Mahajan and Joshi, 2008
τ_s	0.004 MPa from Mahajan and Joshi, 2008
G_I, G_{II}	0.35, 0.50 J/m^s from Mahajan and Joshi, 2008
v_s, v_p, ρ	150, 332 m/s from Guillemot et al., 2021 (converted to μ, λ, E)
r_{mode}	User input (0 = Mode I, 1 = Mode II)

3.2 Source Generation

Results

4.1 Preliminary Results & Discussion

The following results shown are for the supershear and sub-rayleigh case of the custom stfs. The sources are spaced 0.1m apart between 30m and 270m and have a ricker wavelet in the xx, a morlet in the yy and a gaussian in the xy direction. The xx and xy direction forces are positive with magnitude 1, the xy direction is negative in -1.

4.1.1 Frequency-Wavenumber (fk) Plot

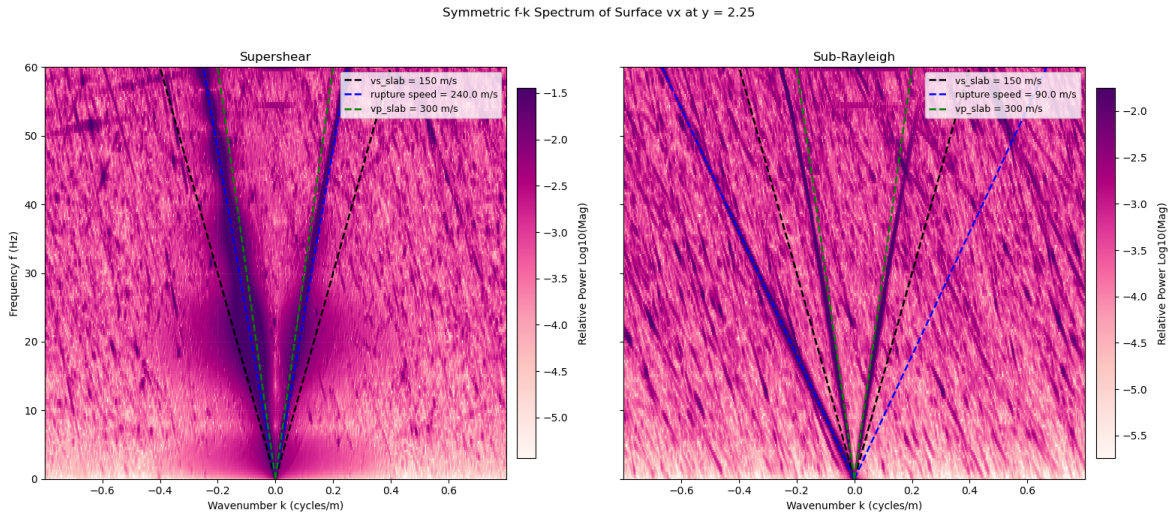


Figure 4.1: F-k plot for the supershear (left) and sub-rayleigh (right) case. Slab p-wave velocity in green, s-wave velocity in black and respective rupture speed in blue. Colorbar shows relative energy (log10 of the magnitude of energy).

When comparing the supershear and the sub-rayleigh fk-plots in figure ??, it becomes clear that the energy is distributed differently in the two cases. The supershear case shows a concentration of energy between the rupture speed (blue line) and the vp (green line) of the slab on both sides of the spectrum. The sub-rayleigh case on the other hand shows three concentrations of energy, two around the positive and negative vp of the slab and one at the negative rupture speed. In all cases, the energy

is more concentrated around the vp and or rupture speed for lower frequencies, at higher frequencies, the concentration of energy starts to deviate from speeds present in the medium in figure ??.

In the supershear case, the dark spots of concentrated energy are more narrow on the positive side compared to the negative side, where they are broader and more smeared out. In the sub-rayleigh case on the right side of figure ??, the dark lines of concentrated energy are much more narrow and distinct.

Figure ?? also clearly shows the distinction in where the energy concentrates. In the sub-rayleigh case, as described above, the energy is concentrated around the p-wave speed on both sides and the rupture speed only in negative direction. In supershear crack propagation, the energy is concentrated between vp and the rupture speed.

4.1.2 Discussion of fk-plot

The narrower energy band on the positive side of the spectrum, as observed in the supershear case in figure ?? can be explained by source directivity. ben-menahemRadiationSeismicSurfacewaves1961 derive directionality factors for seismic waves propagating on either side of a fault. The directionality factor is

$$D = \left| \frac{\left(\frac{C}{v} + \cos \theta_0\right) \sin \left[\frac{\pi b}{\lambda} \left(\frac{C}{v} - \cos \theta_0\right) \right]}{\left(\frac{C}{v} - \cos \theta_0\right) \sin \left[\frac{\pi b}{\lambda} \left(\frac{C}{v} + \cos \theta_0\right) \right]} \right| \quad (4.1)$$

(ben-menahemRadiationSeismicSurfacewaves1961). Analogous to the case described here, where the crack starts at one point in the medium, where the energy travels in the direction of the crack but also opposite to the direction of crack propagation. ben-menahemRadiationSeismicSurfacewaves1961 describe that energy is concentrated in the direction of propagation, so in the positive direction. When looking at the directivity function in equation ?? at 0 degrees (direction of propagation), the function becomes

$$D = \left| \frac{\left(\frac{C}{v} + 1\right) \sin \left[\frac{\pi b}{\lambda} \left(\frac{C}{v} - 1\right) \right]}{\left(\frac{C}{v} - 1\right) \sin \left[\frac{\pi b}{\lambda} \left(\frac{C}{v} + 1\right) \right]} \right| \quad (4.2)$$

, so if the phase velocity (C) is close to the rupture velocity (v), this can lead to a concentration of energy. The opposite is true in the negative direction (180 degrees), where the amplitude is reduced. This explains why the energy is more concentrated and less smeared out in the positive direction compared to the negative direction.

The concentration between vp and the rupture speed in the supershear case can be explained by the radiation of energy. The rupture velocity in the medium exceeds the shear wave speed, so the s-waves no longer decay, they radiate away from the fault as Mach cones (dunhamNearSourceGroundMotion2005). These waves have a certain angle which is described by

$$\xi = \sin^{-1} \left(\frac{c_s}{v} \right) = \tan^{-1} \left(\frac{1}{\hat{\eta}_s} \right) \quad (4.3)$$

, so is determined by the ratio between the rupture speed and the s-wave speed (dunhamNearSourceGroundMotion2005). These waves are tied to the rupture front, which can explain why the energy is concentrated between the p-wave speed and the rupture speed in the supershear case in figure ?? . The p-wave speed in this case is still faster than the rupture, so this energy outruns the rupture, which means the p-waves precede. The rupture and the Mach cone associated with it arrive second, which explains why the energy is concentrated between the two, according to rosakisIntersonicShearCracks2002.

In the sub-rayleigh case, the energy is concentrated at the p-wave speeds, which are the fastest speeds present in the medium. The energy is dominated by the starting and stopping phases, which are radiated by the abrupt starting and stopping of the rupture (madariagaHighfrequencyRadiationCrack1977).

These are governed by the material speeds, so the p- and s-wave speeds (madariagaHighfrequencyRadiationCrack1977). The energy concentration observed at the negative rupture speed could be explained by evanescence of the waves (dunhamNearSourceGroundMotion2005), which means that the amplitude of the waves decays exponentially with their distance to the fault. The furthest distance from the end of the fault is opposite to the direction of propagation, which means that the p- and s-wave energy has decayed the most there. This could allow energy from the fault to show up in the f_k plot in figure ?? at that location because it is no longer overshadowed by the stronger energy of the propagating waves. This could also explain why this energy is only observed in the negative direction.

The strong contrast in how the energy is concentrated between the sub-rayleigh and supershear case in ?? can be explained by the number of locations the energy is concentrated. As described above, the supershear regime has two energy fronts, whereas the sub-rayleigh case has only one. This explains why the energy bands in the supershear case are much more broad compared to the sub-rayleigh case.

Discussion

5.1 Section Title

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Outlook

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Conclusion

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The subfault onset time is given by t_o .

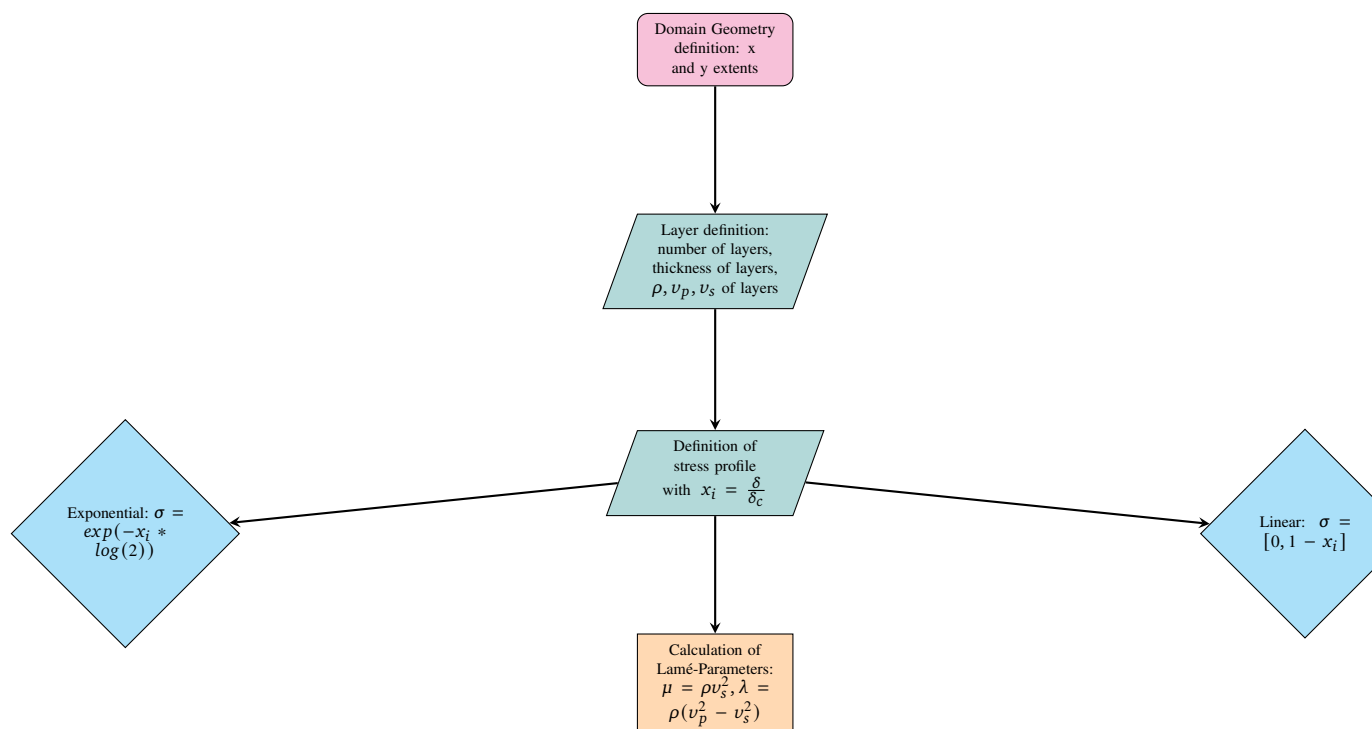
Horizontal coordinates match x .

Code Repository

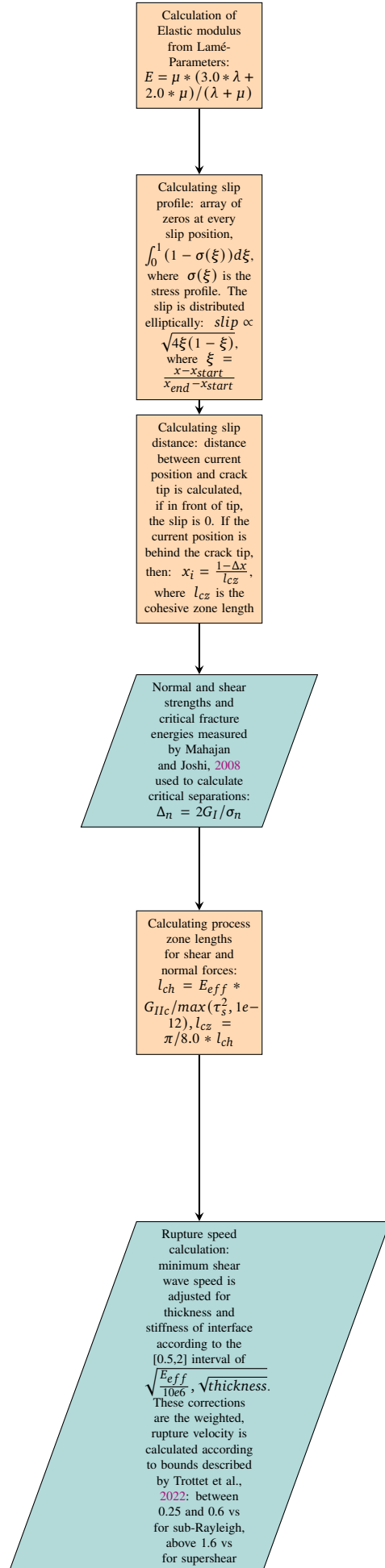
The complete code of all models described can be found on github: <https://github.com/s-bachmann/Master-Thesis>.

Code Workflow

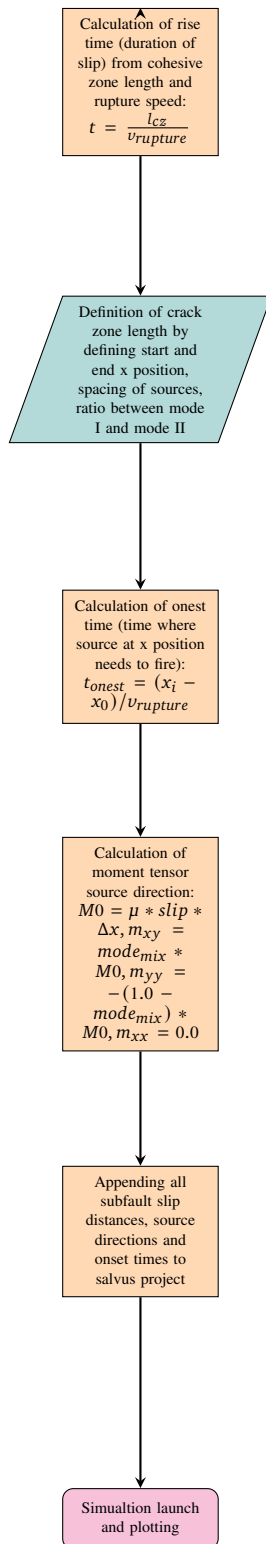
Part 1: Domain and Stress Profile Setup



Part 2: Material Properties and Process Zone Calculation



Part 3: Rupture Propagation and Source Definition



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